the principle of relativity: since all force-free motion is relative, it has meaning only by comparison with other objects or individuals also undergoing force-free motion. There is no way for you to determine anything about your state of motion without making some direct or indirect comparison with “outside” objects. There simply is no notion of “absolute” constant-velocity motion; only comparisons have any physical meaning.

In fact, Einstein realized that the principle of relativity makes an even grander claim: the laws of physics—whatever they may be—must be absolutely identical for all observers undergoing constant-velocity motion. If George and Gracie are not just floating solo in space, but, rather, are each conducting the same set of experiments in their respective floating space-stations, the results they find will be identical. Once again, each is perfectly justified in believing that his or her station is at rest, even though the two stations are in relative motion. If all of their equipment is identical, there is nothing distinguishing the two experimental setups—they are completely symmetric. The laws of physics that each deduces from the experiments will likewise be identical. Neither they nor their experiments can feel—that is, depend upon in any way—constant-velocity travel. It is this simple concept that establishes complete symmetry between such observers; it is this concept that is embodied in the principle of relativity. We shall shortly make use of this principle to profound effect.

The Speed of Light

The second key ingredient in special relativity has to do with light and properties of its motion. Contrary to our claim that there is no meaning to the statement “George is traveling at 10 miles per hour” without a specified benchmark for comparison, almost a century of effort by a series of dedicated experimental physicists has shown that any and all observers will agree that light travels at 670 million miles per hour regardless of benchmarks for comparison.

This fact has required a revolution in our view of the universe. Let’s first gain an understanding of its meaning by contrasting it with similar statements applied to more common objects. Imagine it’s a nice, sunny day and you go outside to play a game of catch with a friend. For a while, you both leisurely throw the ball back and forth